

National University of Computer and Emerging Sciences, Lahore Campus



Course Name:	Physics for Engineers	Course Code:	NS1002
Program:	BS(Electrical Engineering)	Semester:	Fall 2022
Duration:	3 Hours	Total Marks:	100, Wt 50%
Paper Date:	17-12-2022	Exam Type:	FINAL (Obj+Sub)
Section:	All	Number of Pages:	7

Name: _____ Roll No. _____ Section: _____

Instruction/Notes:

1. Attempt all Questions.
2. Please write your answers within the space provided.
3. You can use rough sheet, but that won't be marked.

constants: $g=9.8 \text{ m/s}^2$; $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 \cdot \text{N}^{-1} \cdot \text{m}^{-2}$; $e = \text{charge of electron/proton} = 1.60 \times 10^{-19} \text{ C}$
 Speed of Light $= 3 \times 10^8 \text{ m/s}$; mass of proton $= 1.67 \times 10^{-27} \text{ kg}$; mass of electron $= 9.11 \times 10^{-31} \text{ kg}$;
 $\mu_0 = 4\pi \times 10^{-7} \text{ T.m/A}$

Question No.	1 (Objective)	2	3	4	Total
Maximum Marks	10	30	30	30	100
Marks Obtained					

Question 1: Mention only one correct option. [10 marks] [CLO: 02]

1) An electroscope is charged by induction using a glass rod that has been made positive by rubbing it with silk. The electroscope leaves:

- A) gain electrons
 B) gain protons
 C) lose electrons
 D) lose protons
 E) gain an equal number of protons and electrons

2) Two small charged objects repel each other with a force F when separated by a distance d . If the charge on each object is reduced to one-fourth of its original value and the distance between them is reduced to $d/2$ the force becomes:

- A) $F/16$
 B) $F/8$
 C) $F/4$
 D) $F/2$
 E) F

3) Particle 1 with charge q_1 , and particle 2, with a charge q_2 , are on the x axis, with particle 1 at $x = a$ with and particle 2 at $x = -2a$. For the net force on a third charged particle, at the origin to be zero q_1 and q_2 must be related by $q_2 =$:

- A) $2q_1$
 B) $4q_1$
 C) $-2q_1$
 D) $-4q_1$
 E) $-q_1/4$

4) A particle with charge $2\mu\text{C}$ is placed at the origin. An identical particle, with the same charge, is placed 2 m from the origin on the x axis, and a third identical particle, with the same charge, is placed 2 m from the origin on the y axis. The magnitude of the force on the particle at the origin is:

- A) $9.0 \times 10^{-3} \text{ N}$
 B) $6.4 \times 10^{-3} \text{ N}$
 C) $1.3 \times 10^{-2} \text{ N}$
 D) $1.8 \times 10^{-2} \text{ N}$
 E) $3.6 \times 10^{-2} \text{ N}$

5) Two identical charges, 2.0 m apart, exert forces of magnitude 4.0 N on each other. The value of either charge is:

- A) $1.8 \times 10^{-9} \text{ C}$
 B) $2.1 \times 10^{-5} \text{ C}$
 C) $4.2 \times 10^{-5} \text{ C}$
 D) $1.9 \times 10^3 \text{ C}$
 E) $3.8 \times 10^5 \text{ C}$

6) The electric field at a distance of 10 cm from an isolated point particle with a charge of $2 \times 10^{-9} \text{ C}$ is:

- A) 1.8 N/C
 B) 18 N/C
 C) 180 N/C
 D) 1800 N/C
 E) none of these

7) An isolated charged point particle produces an electric field with magnitude E at a point 2 m away. At a point 1 m from the particle the magnitude of the field is:

- A) E
 B) $2E$
 C) $4E$
 D) $E/2$
 E) $E/4$

8-B
 9-A
 10-E

- 8) An electric dipole consists of a particle with a charge of $+6 \times 10^{-6} \text{ C}$ at the origin and a particle with a charge of $-6 \times 10^{-6} \text{ C}$ on the x axis at $x = 3 \times 10^{-3} \text{ m}$. Its dipole moment is:
- A) $1.8 \times 10^{-3} \text{ C}\cdot\text{m}$, in the positive x direction
 B) $1.8 \times 10^{-3} \text{ C}\cdot\text{m}$, in the negative x direction
 C) $0 \text{ C}\cdot\text{m}$, because the net charge is 0
 D) $1.8 \times 10^{-3} \text{ C}\cdot\text{m}$, in the positive y direction
 E) $1.8 \times 10^{-3} \text{ C}\cdot\text{m}$, in the negative y direction
- 9) A spherical shell has an inner radius of 3.7 cm and an outer radius of 4.5 cm. If charge is distributed uniformly throughout the shell with a volume density of $6.1 \times 10^{-4} \text{ C/m}^3$ the total charge is:
- A) $1.0 \times 10^{-7} \text{ C}$
 B) $1.3 \times 10^{-7} \text{ C}$
 C) $2.0 \times 10^{-7} \text{ C}$
 D) $1.7 \times 10^{-6} \text{ C}$
 E) $5.0 \times 10^{-6} \text{ C}$
- 10) A cylinder has a radius of 2.1 cm and a length of 8.8 cm. Total charge $6.1 \times 10^{-7} \text{ C}$ is distributed uniformly throughout. The volume charge density is:
- A) $7.4 \times 10^{-11} \text{ C/m}^3$
 B) $5.3 \times 10^{-5} \text{ C/m}^3$
 C) $5.3 \times 10^{-5} \text{ C/m}^2$
 D) $8.5 \times 10^{-4} \text{ C/m}^3$
 E) $5.0 \times 10^{-3} \text{ C/m}$

Question 2 [30 marks] (CLO: 03)

2(i): Derive a mathematical expression between current density and drift velocity of conduction electrons.

$$q = nALxe$$

$$t = \frac{L}{v_d} = t$$

$$i = \frac{q}{t} = \frac{nALe}{L/v_d} = \frac{J}{ne}$$

$$\boxed{\vec{J} = ne\vec{v}_d}$$

2(ii): Differentiate between Ohmic and Non-ohmic conductors with examples.

2(iii): Derive two alternate forms of Ohm's law.

$$V = iR \quad (1)$$

$$i = JA \quad (2)$$

$$R = \frac{\rho L}{A} = \frac{L}{\sigma A} \quad (3) \text{ Put in 1}$$

$$\frac{1}{\rho} = \sigma ; V = JA \times \frac{L}{\sigma A}$$

$$V = \frac{JL}{\sigma}$$

$$\vec{J} = \frac{\sigma V}{L} = \sigma \vec{E}$$

$$\vec{J} = \sigma \vec{E}$$

$$J = \frac{1}{\rho} E$$

$$\boxed{E = \rho J}$$

$$\boxed{\rho = \frac{E}{J}}$$

2(iv): What will be the resistivity of a metal wire of 2 m in length and 0.6 mm in diameter, if the resistance of the wire is 50 Ω ?

$$R = 50 \Omega, \text{ length} = 2 \text{ m}, \text{ dia} = 0.6 \text{ mm}.$$

$$r = 0.3 \text{ mm} = 3 \times 10^{-4} \text{ m}. \quad \rho = ?$$

$$A = \pi r^2 = 3.14 \times (3 \times 10^{-4})^2 = 2.826 \times 10^{-8} \text{ m}^2$$

$$= 2.826 \times 10^{-7} \text{ m}^2$$

$$\rho = \frac{RA}{l} = \frac{50 \times 2.826 \times 10^{-7}}{2} = 25 \times 2.826 \times 10^{-7} \Omega \cdot \text{m}$$

$$\boxed{\rho = 7.065 \times 10^{-6} \Omega \cdot \text{m}}$$

1 CLO

2(v): The 12-gauge copper wire in a home has a cross-sectional area of $3.31 \times 10^{-6} \text{ m}^2$ and carries a current of 10 A. Calculate the magnitude of the electric field in the wire. Assume the resistivity of copper: $1.72 \times 10^{-8} \Omega \cdot \text{m}$.

$$E = \rho J = \rho \frac{i}{A} = \frac{1.72 \times 10^{-8} \times 10}{3.31 \times 10^{-6}}$$

$$E = 5.2 \times 10^{-2} \text{ V/m}$$

1 CLO

Question 3 [30 marks] (CLO: 03)

3(i): Consider a copper strip carrying a current that is immersed in a magnetic field. Derive an expression of the potential developed across the width of the strip. Draw an illustration as well.

$$F_E = F_B$$

$$eE = ev_d B$$

$$E = v_d B \quad (1)$$

$$J = ne v_d$$

$$\frac{J}{ne} = v_d$$

$$J = \frac{i}{A}$$

$$v_d = \frac{i}{Ane}$$

$$A = lnd$$

$$v_d = \frac{i}{lndne}$$

$$V = Ed \quad \Rightarrow \quad E = \frac{V}{d} \quad (2)$$

$$E = v_d B \quad (1)$$

$$\frac{V}{d} = \frac{i}{Ane} B$$

$$\boxed{V = \frac{iB}{lne}}$$

$$\boxed{n = \frac{Bi}{Vle}}$$

$$\frac{Bi}{Vle} = n$$

2 CLO

3(ii) A beam of electrons is projected into a chamber by an electron gun. The electrons enter the plane of the page with speed v and then move in a region of uniform magnetic field B directed out of the plane. Derive expressions of all parameters (radius of the circular path, time period, frequency, and angular frequency) that characterize the circular motion of the electrons.

$$qvB = \frac{mv^2}{r}$$

$$r = \frac{mv}{qB} \quad \checkmark$$

$$T = \frac{2\pi r}{v} = \frac{2\pi}{v} \times \frac{mv}{qB} = \frac{2\pi m}{qB} \quad \checkmark$$

$$f = \frac{qB}{2\pi m} \quad \checkmark, \quad \omega = 2\pi f = \frac{qB}{m} \quad \checkmark$$

3(iii): What are the three types of magnetism? Explain them briefly.

- Intrinsic mag. field
- Electromagnetism
- Permanent magnets

3(iv): A straight, horizontal length of copper wire has current $i = 28 \text{ A}$ through it. What are the magnitude and direction of the minimum magnetic field needed to suspend the wire—that is, to balance the gravitational force on it? The linear density (mass per unit length) of the wire is 46.6 g/m .

$$F_B = F_g$$

$$iLB = mg$$

$$B = \frac{mg}{iL} = \frac{(m/L)g}{i}$$

$$= \frac{46.6 \times 10^{-3} \times 9.8}{28}, \quad B = 1.6 \times 10^{-2} \text{ T}$$

(horizontal & Rightwards)

1 CLO

Question 4 [30 marks] (CLO: 03)

4(i) Calculate induced emf, induced current, and the rate at which thermal energy appears in the loop which is being pulled out of a magnetic field at a constant speed. While the loop is moving, a clockwise current is induced into the loop.

$$\mathcal{E} = \frac{d\Phi_B}{dt} = \frac{d}{dt}(BA) = \frac{d}{dt}(BLx)$$

$$= BL \frac{dx}{dt}$$

$$\mathcal{E} = BLv$$

$$i = \frac{V}{R} = \frac{\mathcal{E}}{R} = \frac{BLv}{R}$$

$$F = iL \times B = iLB = \frac{BLv}{R} B$$

$$F = \frac{B^2 L^2 v}{R}$$

$$P = Fv = \frac{B^2 L^2 v^2}{R}$$

1 CLO

4(ii) Describe the Reformulation of Faraday's law when a particle of charge q_0 moves around the circular path (see fig. below). Calculate induced emf expression as well.

$$W = \int \vec{F} \cdot d\vec{s} = (q_0 E)(2\pi r)$$

$$\mathcal{E} = 2\pi r E$$

$$W = \oint \vec{F} \cdot d\vec{s} = q_0 \oint \vec{E} \cdot d\vec{s}$$

$$\mathcal{E} = \oint \vec{E} \cdot d\vec{s}$$

$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

$$\oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$$

1000

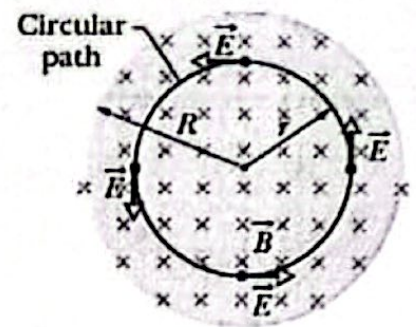


Figure explanation:

If the magnetic field increases at a steady rate, a constant induced current appears, as shown, in the copper ring of radius r .

An induced electric field exists even when the ring is removed; the electric field is shown at four points.

4(iii): Explain Faraday's law for the induced emf from electromagnetism.

1000

4(iv): A coil formed by wrapping 50 turns of wire in the shape of a square is positioned in a magnetic field so that the normal to plane of the coil makes an angle of 30° with the direction of the field. When the magnetic field is increased uniformly from $200 \mu\text{T}$ in 0.4 s , an emf of magnitude 80 mV is induced in the coil. What is the total length of the wire? [Assume that this square coil has some reasonable size between 1 cm and 1 m across, then the total length of wire would be between 2 m and 200 m .]

$$\mathcal{E} = -N \frac{d\Phi_B}{dt}, \quad \mathcal{E} = -N \frac{d}{dt} (BA \cos \theta)$$

$$\mathcal{E} = -NA \cos \theta \frac{dB}{dt}$$

$$|\mathcal{E}| = NA \cos \theta \left(\frac{\Delta B}{\Delta t} \right)$$

$$A = \frac{|\mathcal{E}|}{N \cos \theta \left(\frac{\Delta B}{\Delta t} \right)}$$

$$\mathcal{E} = 80 \text{ mV}$$

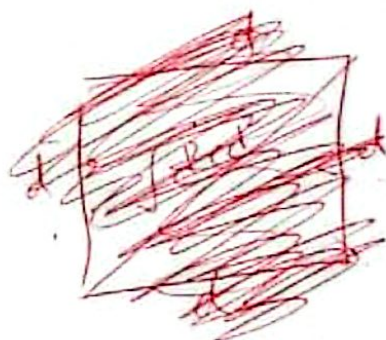
$$N = 50$$

$$\theta = 30^\circ$$

$$\Delta B = 200 \mu\text{T}$$

$$\Delta t = 0.4 \text{ sec}$$

$$\begin{aligned} A &= \frac{80 \times 10^{-3}}{50 \cos 30^\circ \left(\frac{200 \times 10^{-6}}{0.4} \right)} \\ &= \frac{80 \times 10^{-3}}{50 \times 0.866 (500 \times 10^{-6})} \\ &= \frac{80 \times 10^{-3}}{0.02165} \end{aligned}$$



$$L = 4d \times N$$

$$A = 3.695 \text{ m}^2$$

$$L = 4N\sqrt{A} = 4 \times 50 \times \sqrt{3.695}$$

$$L = 384.44 \text{ m}$$

$$d = \sqrt{A}$$

$$A = d^2 = d \times d$$

$$\sqrt{A} = \sqrt{d^2} \quad d = \sqrt{A}$$

Taking sq root both sides